Bounded Excess Entropy for Adaptive Binary Simulation

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Abstract

Consider a simulator that draws one random seed and subsequently answers every adaptive sequence of actions deterministically. Each action specifies a probability law, and the simulator must reproduce the prescribed conditional distributions exactly under every deterministic adaptive policy. We show that, for any fixed finite family of strictly positive binary laws, the minimum seed entropy satisfies $C_n = nh_{\max} + \gamma + o(1)$, with $0 \le \gamma < \infty$. The excess is nondecreasing, and its limit vanishes precisely when all available laws agree up to exchanging their two outcomes. The argument combines an exact invariance under action-dependent output relabeling with the strict-entropy-gap theorem of the accompanying repository. More generally, the conclusion holds whenever all entropy-maximizing actions are equivalent under bijective output relabeling.

1 Exact causal simulation

Let A be a finite nonempty action set. Each action $a \in A$ has a finite output alphabet Y_a and a strictly positive probability law K_a.

At round t, a deterministic policy selects an action using the preceding action–output history. The prescribed conditional law of the next output is K_a. The same family of laws is available at every history.

A causal simulator consists of a finite random seed S and deterministic response functions

$$Y_t = f_t(S, H_{t-1}, A_t).$$

One seed distribution and one decoder must work for every deterministic adaptive policy. The decoder depends on the observed history and current action, rather than on the identity or future choices of the policy.

Write $C_n(K)$ for the infimum of $H(S)$ over exact simulators through n rounds, and set

$$h_{\max} = \max_{a \in A} H(K_a).$$

All entropies are measured in bits. Repeating an entropy-maximizing action gives the elementary lower bound $C_n(K) \ge nh_{\max}$.

2 Relabeling invariance

Two actions a, b are *equivalent* if there is a bijection

$$\phi : Y_a \longrightarrow Y_b, \qquad K_b(\phi(y)) = K_a(y).$$

Let $\bar{K}$ contain one representative from each equivalence class.

Lemma 1. *For every finite horizon,*

$$\boxed{C_n(K) = C_n(\bar{K}).}$$

Proof. Restricting a simulator to representative actions gives $C_n(\bar{K}) \le C_n(K)$.

Conversely, fix a representative $r(a)$ and a probability-preserving bijection $\phi_a : Y_{r(a)} \to Y_a$ for every action a. Given a simulator for $\bar{K}$, simulate action a by querying $r(a)$ and returning ϕ_a applied to its output. Convert each preceding action–output pair (a, y) into

$$(r(a), \phi_a^{-1}(y))$$

before supplying the history to the representative decoder.

To verify adaptive exactness, fix an original policy. Along a representative history, reconstruct the original history recursively: the original policy determines the next action, and its associated bijection converts the representative output into the original output. This defines a deterministic adaptive policy on the representative system; its values outside the recursively consistent histories may be chosen arbitrarily. Its exact transcript law pushes forward to the prescribed original transcript law.

The constructed decoder uses only the seed, history, and current action, so shared-history consistency is preserved. No additional randomness is introduced. Thus $C_n(K) \le C_n(\bar{K})$. □

3 A finite limiting excess

We use the following established result from the accompanying development [1]:

> **Strict entropy gap theorem.** If a stationary finite positive kernel has exactly one entropy-maximizing action, then $C_n \le nh_{\max} + B$ for a finite constant B independent of n.

Theorem 2. *Suppose all entropy-maximizing actions of K are equivalent under bijective output relabeling. Then there is a finite $\gamma \ge 0$ such that*

$$\boxed{C_n(K) = nh_{\max} + \gamma + o(1).}$$

Moreover, $C_n(K) - nh_{\max}$ is nondecreasing.

Proof. After passing to representatives, exactly one action maximizes entropy. Lemma 1 and the strict entropy gap theorem imply

$$0 \le C_n(K) - nh_{\max} \le B.$$

For monotonicity, take any exact seed S for $n+1$ rounds and choose an entropy-maximizing action first. Let Y be its output. For each y, the conditional seed $S \mid Y = y$, together with the decoder restricted to that history, is an exact seed for the remaining n rounds. Indeed, any continuation policy can be prefixed by that first action, and stationarity identifies its continuation problem with the original n-round problem.

Since Y is a deterministic function of S,

$$H(S) = H(Y) + H(S \mid Y) \ge h_{\max} + C_n(K).$$

Taking the infimum gives $C_{n+1}(K) \ge h_{\max} + C_n(K)$. The excess is therefore nonnegative, nondecreasing, and bounded. It converges to a finite limit γ. □

4 Complete binary classification

Corollary 3. *Let $K_a = \text{Bernoulli}(p_a)$, where $0 < p_a < 1$. For every fixed finite nonempty action family,*

$$\boxed{C_n = nh_{\max} + \gamma + o(1), \qquad 0 \le \gamma < \infty.}$$

For any entropy-maximizing action a_*,

$$\boxed{\gamma = 0 \iff p_a \in \{p_{a_*}, 1 - p_{a_*}\} \quad \textit{for every } a.}$$

In the zero case, $C_n = nh_{\max}$ at every horizon. Otherwise,

$$0 < C_1 - h_{\max} \le C_n - nh_{\max} \le \gamma \qquad (n \ge 1).$$

Proof. Binary entropy is symmetric under $p \mapsto 1 - p$ and strictly increasing on $(0, \frac{1}{2}]$. Hence all maximizing binary laws are equivalent, and Theorem 2 applies.

If every law is the maximizing law or its complement, draw n independent samples from that law and complement each sample when required by the chosen action. This realizes every adaptive policy with seed entropy $nh_{\max}$.

Conversely, suppose $\gamma = 0$. Monotonicity and nonnegativity imply $C_1 = h_{\max}$. At one round, minimizing seed entropy is equivalent to minimizing the joint entropy of one output for each action: a seed induces such a tuple as a deterministic function, and any such tuple can itself serve as a seed. The minimum is attained because the finite coupling polytope is compact and entropy is continuous.

Let $X = (X_a)_{a \in A}$ be a minimizing coupling. Then

$$H(X) = H(X_{a_*}), \qquad H(X \mid X_{a_*}) = 0.$$

Every X_a is consequently a deterministic function of X_{a_*}. Strict positivity excludes constant functions. A nonconstant function between two binary alphabets is either the identity or the exchange of outcomes. This gives the claimed characterization. If that condition fails, $C_1 > h_{\max}$, and the final inequalities follow from monotonicity and convergence. □

5 Application to reset qubit experiments

Prepare a fresh qubit in $|0\rangle$, evolve it under

$$\mathcal{H} = \frac{E}{2}\sigma_x,$$

and measure in the computational basis after an action-selected duration t_a. The probability of outcome 1 is

$$p_a = \sin^2\left(\frac{Et_a}{2\hbar}\right).$$

For any fixed finite collection of durations satisfying $0 < p_a < 1$, assuming the prescribed reset experiment at every round, Corollary 3 applies to the classical measurement records:

$$C_n = n \max_a H_2(p_a) + \gamma + o(1).$$

Thus adaptive selection among these experiments incurs a bounded excess seed entropy. A growing logarithmic or square-root correction is absent in this model. Equal outcome entropies give exact equality $C_n = nh_{\max}$.

This is a statement about exact classical simulation of the specified records. It does not require or establish a deterministic physical substrate for quantum mechanics.

Verification and scope

The strict entropy gap theorem and residual monotonicity are provided in [1], principally in `CausalSeed/GapBound.lean` and `CausalSeed/GapImage.lean`. This note supplies the mathematical relabeling argument and its binary consequence; it does not claim a completed Lean formalization of those additional statements or an independent rebuild of the repository.

The constant γ is characterized by convergence, not evaluated explicitly. The result concerns a fixed finite action family; no uniform bound is asserted as that family varies. No claim of historical priority is made.

References

[1] J. Mikulík. *One Seed for Every Adaptive Policy.* Mathematical development and Lean sources. Repository consulted 19 October 2026. `https://github.com/jendamikulik/finite-Shannon-theorem`.